Percentage of quantities

1 Complete the table of fraction to percentage conversions.

Fraction	$\frac{1}{1}$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{5}$	$\frac{1}{8}$		$\frac{1}{20}$		$\frac{1}{50}$	$\frac{1}{100}$
Percentage		50%				10%		4%		1%

2 Write the fraction or number that completes the following statements:

a 25% of a quantity is the same as ⬚ of that quantity.

b 200% of a quantity is the same as ⬚ of that quantity.

3 Find the following quantities by first converting the percentage into a fraction.

a 15% of 160 **b** 250% of 12 **c** 16% of 275 **d** 175% of 20

4 Find the following quantities by first converting the percentage into decimal.

a 90% of 20 **b** 220% of 10 **c** 60% of 60 **d** 21% of 400

5 Show a way to correctly calculate 25% of 140.

6 Find the following:

a 10% of 90 **b** 20% of 45 **c** 1% of 900 **d** 2% of 400

e 500% of 11 **f** 350%of 16 **g** 60% of 30 **h** 45% of 160

i 200% of 13 **j** 150%of 18 **k** 80% of 40 **l** 50% of 320

7 Find the following:

a 10% of 80 g **b** 20% of \$60 **c** 30% of \$200 **d** 2% of 300 mL

e 90% of 80 kg **f** 70%of 20 km **g** 25% of 120 L **h** 75% of \$24

8 Follow the steps in each case to determine the given quantity:

a 75% of 28.

i How many groups of 25% go into 75%?

ii What is 25% of 28?

iii Hence, find 75% of 28.

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b 140% of 35.

- **i** How many groups of 20% go into 140%?
- **ii** What is 20% of 35?
- **iii** Hence, find 140% of 35.

c 28% of of 175.

- **i** How many groups of 4% go into 28%?
- **ii** What is 4% of 175?
- **iii** Hence, find 28% of 175.

d 7% of 300.

- **i** How many groups of 1% go into 7%?
- **ii** What is 1% of 300?
- **iii** Hence, find 7% of 300.

Percentage comparisons

9 Which is greater:

a 62% of 38 or 62% of 46?

b 27% of 230 or 13% of 230?

c 70% of 30 or 250% of 18?

d 45% of 180 or 7% of 700?

e 13% of 17 or 29% of 13?

f 36% of 48 or 24% of 63?

10

a When finding a percentage of 82 that is above 100%, will the result be smaller or larger than 82?

b When finding a percentage of 82 that is below 100%, will the result be smaller or larger than 82?

c Find the following:

- **i** 60% of 82
- **ii** 120% of 82

Applications

11 Lisa scored 95% on her geography exam, which was marked out of 160. What was Lisa's actual mark out of 160?

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12 A theatre has a capacity of 3000 people. On the opening night the theatre was 97% full. On the second night it was 78% full.

a How many people attended the theatre on the first night?

b How many people attended theatre second night?

c How many people attended the theatre in total during the first 2 nights.

13 A stadium can hold 5000 people. The stadium is 70% full and 11% of the audience are children.

a How many people are there in the stadium?

b How many children are there in the stadium?

14 Yuri is eating a box of cookies. The nutritional information states that each cookie contains 7% of his recommended daily intake of sugar.

If his recommended daily intake of sugar is $37\,500$ mg, how many milligrams of sugar does each cookie contain?

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